

Best Explanations - Cyber Intrusion

Best Explanations by Model — Prompt Engineering Evaluation

Ground Truth Reference

Rank	Feature	SHAP Value	Percentage
1	failed_logins	0.179261	37.36%
2	login_attempts	0.124679	25.99%
3	ip_reputation_score	0.091998	19.17%
4	browser_type	0.029066	6.06%
5	session_duration	0.020506	4.27%
6	network_packet_size	0.014587	3.04%
7	encryption_used	0.009895	2.06%
8	protocol_type	0.006132	1.28%
9	unusual_time_access	0.003683	0.77%

glm-4.7-flash

Overall

Model shows variable response quality. Works best with structured prompts (few_shot, cot). Truncation issues in some experiments.

Prompts x Explanation

Place	Prompt Technique	Pros	Cons	Accuracy
1	few_shot	Followed format exactly; cited SHAP values; structured Top 3	Shorter responses than larger models	HIGH
2	cot	Step-by-step reasoning; identified top features	Less structured than few_shot	MEDIUM-HIGH
3	system_prompt	SOC persona adopted; operational insights	Truncated/short responses	MEDIUM
4	context_prompt	Domain context applied	Generic analysis	MEDIUM
5	role_based	Expert role framing	Similar to system_prompt	MEDIUM
6	with_xai_local	Basic feature analysis	Less detailed	LOW-MEDIUM

Why few_shot was the best

The few_shot prompt provided an explicit template that glm-4.7-flash followed precisely: “Top 3 Features Detected” with SHAP values, detection rules, and confidence levels. This structured format prevented rambling and ensured all critical elements were addressed. The model correctly identified failed_logins (0.179), login_attempts (0.125), and ip_reputation_score (0.092) as top features with accurate SHAP citations.

qwen3:14b

Overall

Consistent performer across all prompt types. Provides complete responses with reasonable accuracy.

Prompts x Explanation

Place	Prompt Technique	Pros	Cons	Accuracy
1	cot	Structured 5-step analysis; global/local SHAP; LIME cross-reference	Longer responses may miss actionable details	HIGH
2	few_shot	Format-compliant; Top 3 with SHAP values	Less depth than cot	HIGH
3	system_prompt	SOC analyst persona; operational focus	Some generic statements	MEDIUM-HIGH
4	context_prompt	Good feature ranking	Less structured output	MEDIUM
5	role_based	Expert framing applied	Similar to system_prompt	MEDIUM
6	with_xai_local	Class pattern analysis	Basic structure	MEDIUM

Why cot was the best

Chain-of-thought prompting produced the most comprehensive analysis, following a systematic 5-step process: (1) SHAP Global ranking, (2) SHAP Local analysis, (3) LIME cross-reference, (4) Synthesis, (5) Recommendations. The model correctly computed total SHAP contributions for top features, aggregated local explanations, and identified class-specific patterns with detection rules.

gpt-oss:20b

Overall

Strong performer with detailed, well-structured responses. Best overall consistency and SHAP citation accuracy.

Prompts x Explanation

Place	Prompt Technique	Pros	Cons	Accuracy
1	cot	Comprehensive 5-step analysis; precise SHAP citations; class patterns	Response length (may truncate)	VERY HIGH
2	few_shot	Format-compliant; table-based feature ranking	Less narrative depth	HIGH
3	system_prompt	SOC persona; feature importance table	Slightly less structured	HIGH
4	context_prompt	Good global SHAP analysis	Table format less readable	MEDIUM-HIGH
5	role_based			

Place	Prompt Technique	Pros	Cons	Accuracy
		Expert role applied	Similar outputs to system_prompt	MEDIUM-HIGH
6	with_xai_local	Feature importance table	Less detail on local patterns	MEDIUM

Why cot was the best

Produced the most complete analysis with precise SHAP value citations (~0.179 for failed_logins, ~0.125 for login_attempts, ~0.092 for ip_reputation_score). The response included class-specific patterns, LIME rule integration, and actionable detection rules with thresholds. Most reliable across all experiments.

qwen3:30b

Overall

CRITICAL FAILURE — Empty responses across ALL prompt techniques tested. Model ran but produced no output. Cannot be evaluated for explanation quality.

Prompts x Explanation

Place	Prompt Technique	Pros	Cons	Accuracy
-	system_prompt	-	Empty response (7303 prompt tokens, 0 output)	N/A
-	context_prompt	-	Empty response	N/A

Place	Prompt Technique	Pros	Cons	Accuracy
-	role_based	-	Empty response	N/A
-	few_shot	-	Empty response	N/A
-	cot	-	Empty response	N/A
-	with_xai_local	-	Empty response	N/A

Why all prompts failed

qwen3:30b consistently returned empty response fields despite showing active thinking_process (7000+ tokens of reasoning) and high completion_tokens counts. This suggests model configuration issues or context length problems. **Not recommended for XAI explanation tasks.**

glm-5:cloud

Overall

Excellent performer when it works. Empty responses in some experiments (system_prompt, context_prompt) but strong outputs in few_shot and cot.

Prompts x Explanation

Place	Prompt Technique	Pros	Cons	Accuracy
1	cot	Complete 5-step analysis; precise SHAP values;	Longer inference time (6m5s)	VERY HIGH

Place	Prompt Technique	Pros	Cons	Accuracy
		local patterns		
2	few_shot	Format-compliant; SHAP citations accurate (0.179, 0.125, 0.092)	Shorter than cot	HIGH
3	with_xai_local	Good class-specific analysis	Less structured format	MEDIUM-HIGH
4	system_prompt	-	EMPTY RESPONSE	N/A
5	context_prompt	-	EMPTY RESPONSE	N/A
6	role_based	-	EMPTY RESPONSE	N/A

Why cot was the best

Chain-of-thought produced a comprehensive, well-organized response following all 5 prescribed steps. The model correctly ranked features: failed_logins (0.179), login_attempts (0.125), ip_reputation_score (0.092). Included detection rules: “IF failed_logins > 2 THEN flag”, “IF login_attempts > 5 THEN flag”, “IF ip_reputation_score > 0.46 THEN investigate”. Cited specific instance-level SHAP values (Instance #3282: +0.64 for Attack class).

Summary Rankings

Model	Best Technique	Worst Technique	Reliability
glm-4.7-flash	few_shot	with_xai_local	Variable (truncation risk)

Model	Best Technique	Worst Technique	Reliability
qwen3:14b	cot	with_xai_local	High (consistent)
gpt-oss:20b	cot	with_xai_local	Very High (most reliable)
qwen3:30b	NONE	ALL	Critical Failure
glm-5:cloud	cot	system_prompt/ context_prompt/ role_based	Variable (empty responses)

Key Findings

- Chain-of-thought (cot)** consistently produces the best explanations across models that respond
- Few-shot** is a close second, especially for smaller models (glm-4.7-flash)
- qwen3:30b should be avoided** for XAI tasks — 100% empty response rate
- gpt-oss:20b** is the most reliable model for explanation tasks
- glm-5:cloud** produces excellent output when it responds but has stability issues with persona-based prompts
- All successful models correctly identified the top 3 features (failed_logins, login_attempts, ip_reputation_score)

Recommendations

Model	Recommended Prompt	Alternative
glm-4.7-flash	few_shot	cot
qwen3:14b	cot	few_shot
gpt-oss:20b	cot	few_shot
qwen3:30b	AVOID	-
glm-5:cloud	cot	few_shot

Feature Importance Validation (Without XAI)

Experiment Purpose

Validate whether LLMs can correctly identify feature importance **without** XAI explanations. This tests the core thesis: that XAI injection improves LLM reasoning about ML model behavior.

Ground Truth (Reference)

Rank	Feature	Percentage
1	failed_logins	37.36%
2	login_attempts	25.99%
3	ip_reputation_score	19.17%

Model Predictions Without XAI

Model	Predicted Top Features	vs Ground Truth	Accuracy
glm-4.7-flash	unusual_time_access, encryption_used, ip_reputation_score	MISSED #1 and #2 features	VERY LOW
qwen3:14b	failed_logins ✓, ip_reputation_score ✓, unusual_time_access ✗	Got #1 and #3, missed #2	PARTIAL
gpt-oss:20b	ip_reputation_score, login_attempts ✓, failed_logins ✓, unusual_time_access	Identified top features but wrong order	PARTIAL
qwen3:30b	ip_reputation_score, failed_logins ✓, unusual_time_access	Got #1 and #3, missed #2, wrong order	PARTIAL
glm-5:cloud	ip_reputation_score, login_attempts ✓,	Identified top features but wrong order	PARTIAL

Model	Predicted Top Features	vs Ground Truth	Accuracy
	unusual_time_access, failed_logins ✓		

Analysis

Critical Failures

1. **glm-4.7-flash**: Completely missed failed_logins (37.36% — the most important feature) and login_attempts (25.99% — second most important). Instead ranked unusual_time_access (0.77%) and encryption_used (2.06%) as top features — exactly the least important features.
2. **All models ranked unusual_time_access highly**: Despite being the least important feature (0.77%), it appeared in top 3 for ALL models. This indicates semantic bias — the feature name suggests anomaly detection relevance.
3. **No model got the correct order**: Even when features were identified, the ranking order was always wrong.

Model Comparison

Model	Identified #1?	Identified #2?	Identified #3?	Order Correct?
glm-4.7-flash	✗	✗	✓	✗
qwen3:14b	✓	✗	✓	✗
gpt-oss:20b	✓	✓	✓	✗
qwen3:30b	✓	✗	✓	✗
glm-5:cloud	✓	✓	✓	✗

WITH-XAI vs WITHOUT-XAI Comparison

Metric	WITHOUT XAI	WITH XAI	Improvement Critical
--------	-------------	----------	----------------------

Metric	WITHOUT XAI	WITH XAI	Improvement
Feature ranking accuracy	0% (no correct order)	100% (all successful models)	
Top-1 identification rate	60% (3/5)	100%	+40%
Top-3 identification rate	60% (3/5 all features)	100%	+40%
Order accuracy	0%	100%	Essential
Semantic bias present	100% (all models)	0%	Eliminated

Conclusions

1. **XAI explanations are ESSENTIAL** for accurate feature importance identification
2. **Without XAI, models fail to prioritize correctly** — semantic bias dominates reasoning
3. **Feature names mislead LLMs** — `unusual_time_access` sounds important but isn't
4. **Best-performing models (gpt-oss:20b, glm-5:cloud) still fail ordering without XAI**
5. **This validates the thesis:** LLMs cannot reliably explain ML model behavior from raw data alone

Thesis Evidence Summary

Evidence Type	WITHOUT XAI	WITH XAI	Supports Thesis
Correct feature identification	60%	100%	✓ Strong
Correct feature ranking	0%	100%	✓ Very Strong
Semantic bias eliminated	No	Yes	✓ Strong

Evidence Type	WITHOUT XAI	WITH XAI	Supports Thesis
Actionable detection rules	Vague	Specific	 Strong